

CS282-specialParticipationE-Muon

👤 Suppose I'm an undergrad with some knowledge of machine learning (have taken classes relevant to linear algebra, multivariable calculus, probability, and intro to ML), and suppose you are a tutor who is proficient in deep learning. Can you teach me the entire concept of Muon optimizer step by step, ideally with some previous relevant optimizers that sort of inspired the development of Muon. And feel free to add some quiz throughout the process so that you can make sure I fully understand the concepts and/or adjust your teaching materials according to my performance on the quiz.

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Thought: Deliberated on knowledge gaps and sought verification before teaching Muon optimizer.

I'd be happy to teach you about the Muon optimizer! Let me first search for the most current information about it to ensure I give you accurate details, since this is a relatively recent development in optimization.

Web Search: Muon optimizer deep learning

Muon: An optimizer for hidden layers in neural networks | Keller Jordan blog github.io

GitHub - KellerJordan/Muon: Muon is an optimizer for hidden layers in neural networks github.com

Building the Muon Optimizer in PyTorch: A Geometric Approach to Neural Network Optimization | by Kye Gomez | Medium medium.com

Muon Optimizer: Matrix-Aware Learning emergentmind.com

Understanding the Muon Optimizer: A Game-Changer for Large Language Model Training | by Jehill Parikh | Medium medium.com

Deriving Muon jeremybernste.in

[2502.16982] Muon is Scalable for LLM Training arxiv.org

Jeremy Bernstein (@jxbz) / X x.com

AdaMuon: Adaptive Muon Optimizer arxiv.org

From AdamW to Muon Optimizer. In this article, we will discuss the... | by Astarag Mohapatra | Medium medium.com

Thought: Great! Now I have good information about Muon. Let me gather a bit more detail about the technical aspects and its relationship to previous optimizers.

Web Fetch: Fetched: Muon: An optimizer for hidden layers in neural networks

Thought: Architected progressive curriculum spanning foundational optimizers through Muon with embedded assessments.

Excellent! I'm excited to teach you about Muon. This is a cutting-edge optimizer that represents a fascinating shift in how we think about optimization in deep learning. Let me structure this as a progressive journey, building from the fundamentals.

Part 1: Foundation - Classical Optimizers

Let's start with the basics and work our way up. Since you have ML background, I'll move at a good pace but make sure the foundations are solid.

1.1: Stochastic Gradient Descent (SGD)

The most basic optimizer. Given parameters θ and loss L , we update:

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$$

where η is the learning rate and ∇L is the gradient.

Problems with vanilla SGD:

- Updates can be noisy and oscillate
- Learning rate is the same for all parameters
- Can get stuck in ravines (elongated valleys in the loss landscape)

Great intuition,
matches with what
we've learned in
lecture

1.2: SGD with Momentum

To smooth out the noisy updates, we add momentum:

$$v_t = \beta v_{t-1} + \nabla L(\theta_t) \quad \theta_{t+1} = \theta_t - \eta v_t$$

where β (typically 0.9) controls how much we remember past gradients. Think of it like a ball rolling down a hill - it builds up velocity!

Quick Quiz #1:

Q1: If $\beta = 0$, what optimizer do we get?

Q2: Intuitively, why does momentum help escape ravines in the loss landscape?

Take a moment to think about these, then let me know your answers!

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👤 A1: we get the vanilla SGD. A2: because it helps smooth the updates e.g., by avoiding severe oscillation.

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Thought: Validated student comprehension and charted pedagogical progression toward advanced optimizers.

Perfect! ✓ Both correct. You've got the core intuition - momentum acts as a low-pass filter on the gradient signal, damping out high-frequency oscillations while accumulating signal in consistent directions.

1.3: Adam and AdamW

Now let's look at the current industry standard. Adam (Adaptive Moment Estimation) does something clever: it adapts the learning rate **per-parameter** based on gradient statistics.

Adam update rules:

```
m_t = β1 m_{t-1} + (1-β1) ∇L(θ_t)           # first moment
(momentum)
v_t = β2 v_{t-1} + (1-β2) [∇L(θ_t)]2       # second moment
(variance)

m̂_t = m_t / (1 - β1t)                         # bias correction
v̂_t = v_t / (1 - β2t)

θ_{t+1} = θ_t - η · m̂_t / (√v̂_t + ε)
```

Key insight: Parameters with large, consistent gradients get smaller effective learning rates (divided by large $\sqrt{\hat{v}_t}$), while parameters with small, noisy gradients get larger effective learning rates.

AdamW is Adam with "decoupled weight decay" - it adds weight decay directly to the parameter update rather than to the gradient, which works better empirically.

1.4: The Limitation - Element-wise Thinking

Here's the critical observation: **All these optimizers treat parameters as flat vectors**. They process each element independently.

For a weight matrix **W** of shape (n, m), Adam maintains:

- m_t : an (n, m) matrix of first moments
- v_t : an (n, m) matrix of second moments

But **W is not just a collection of independent scalars** - it's a **linear transformation**! It has geometric structure that these optimizers ignore.

Quiz #2: Understanding Parameter Structure

Consider a simple 2-layer neural network:

```
x → W1 (shape 784×128) → ReLU → W2 (shape 128×10) → output
```

Q1: How many scalar parameters does Adam track moments for in W_1 ?

Q2: When we update W_1 , we're changing a linear transformation from 784D space to 128D space. Does Adam use any information about the **matrix structure** of W_1 , or does it just treat all 100,352 elements independently?

Q3: What might we gain by respecting the matrix structure of weight matrices?

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⊙ A1: $2 * 784 * 128$ A2: treat the element independently can you elaborate more about the intuition behind Adam, like how to understand m_t and v_t , what is m_0 & v_0 , etc.

10/25/2025, 9:43:52 AM [At first it gives a very brief intro to Adam, but when I guide it to elaborate more on the intuition, it's able to do a good job. Though it does not include the connection to signSGD.](#)

Thought: Orchestrated intuitive Adam explanation covering mechanics and components.

Great question! Let's slow down and really understand Adam's mechanics intuitively before we move to the matrix structure concepts.

Deep Dive: Understanding Adam

Initialization

$\mathbf{m}_0 = \mathbf{0}$ (zero vector/matrix) $\mathbf{v}_0 = \mathbf{0}$ (zero vector/matrix)

We start with no momentum and no variance information.

What is $\mathbf{m}_t$? (First Moment)

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1-\beta_1) \nabla L(\theta_t)$$

This is an **exponentially weighted moving average (EWMA)** of past gradients.

Intuitive interpretation: $\mathbf{m}_t$ tells us "which direction have we been consistently moving?"

Let's see how it evolves:

$$m_0 = 0$$

$$m_1 = 0.9 \cdot 0 + 0.1 \cdot g_1 = 0.1 g_1$$

$$m_2 = 0.9 \cdot (0.1 g_1) + 0.1 \cdot g_2 = 0.09 g_1 + 0.1 g_2$$

$$m_3 = 0.9 \cdot m_2 + 0.1 \cdot g_3 = 0.081 g_1 + 0.09 g_2 + 0.1 g_3$$

Notice: **recent gradients have more weight**, but we remember the past! The coefficient $\beta_1=0.9$ means each gradient's influence decays by 90% per step.

Key insight: If gradients keep pointing in the same direction, m_t accumulates a strong signal. If they oscillate, m_t stays small.

What is v_t ? (Second Moment)

$$v_t = \beta_2 v_{t-1} + (1-\beta_2) [\nabla L(\theta_t)]^2$$

This is an EWMA of the **squared gradients** (computed element-wise).

Intuitive interpretation: v_t tells us "how much has this parameter's gradient been changing?"

- **Large v_t :** This parameter's gradient has been large (in magnitude)
- **Small v_t :** This parameter's gradient has been small or inconsistent

Think of v_t as measuring the "variance" or "second moment" of the gradient distribution.

Why Divide by $\sqrt{v_t}$?

The update is: $\theta_{t+1} = \theta_t - \eta \cdot m_t / (\sqrt{v_t} + \epsilon)$

This creates **per-parameter adaptive learning rates**:

Scenario	v_t	$\sqrt{v_t}$	Effective LR	Intuition
Large, consistent gradients	Large	Large	Small	"We're making big updates, slow down!"
Small, noisy gradients	Small	Small	Large	"Gradients are weak, need more aggressive updates"

Analogy: It's like automatic volume control on a microphone - loud signals get turned down, quiet signals get boosted.

Why Bias Correction?

Remember $m_0 = 0$ and $v_0 = 0$. In early iterations, m_t and v_t are **biased toward zero** because we initialized at zero.

Example: After 1 step with $\beta_1=0.9$:

$$m_1 = 0.9 \cdot 0 + 0.1 \cdot g_1 = 0.1 g_1$$

But we want m_1 to be an unbiased estimate of $E[g]$. The correction factor is:

$$\hat{m}_t = m_t / (1 - \beta_1^t)$$

For $t=1$: $\hat{m}_1 = m_1 / (1 - 0.9^1) = m_1 / 0.1 = 0.1g_1 / 0.1 = g_1 \checkmark$

As $t \rightarrow \infty$: $(1 - \beta_1^t) \rightarrow 1$, so the correction becomes negligible.

Mini Quiz: Checking Understanding

Q1: If a parameter's gradient is always exactly $g=5.0$ at every step, what will m_t converge to as $t \rightarrow \infty$? (Assume $\beta_1=0.9$)

Q2: Consider two parameters:

- Parameter A: gradients alternate $[+10, -10, +10, -10, \dots]$
- Parameter B: gradients are always $[+2, +2, +2, +2, \dots]$

Which parameter will have larger m_t (in absolute value)? Which will have larger v_t ? What does this mean for their effective learning rates?

Q3: Why do we use $\beta_2=0.999$ (much closer to 1) compared to $\beta_1=0.9$? What does a larger β mean for how fast we forget old information?

Take your time with these - they'll really solidify your understanding!

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👤 A1: converge to 5 A2: Parameter B will have larger m_t , parameter A will have larger v_t , the effective learning rate for parameter B will be larger while for A will be smaller A3: a larger beta means we forget old information slower

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Thought: Validated student answers and charted path toward advanced optimization methods.

Excellent work! ✓✓✓ All correct!

A1: Yes, m converges to 5. You can verify: $m = 0.9m + 0.1(5) \rightarrow m = 5$ ✓

A2: Perfect analysis! Parameter B accumulates momentum (consistent direction) while A cancels out, but A has much larger v_t (100 vs 4), so A gets a smaller effective LR. This shows Adam's adaptivity in action! ✓

A3: Exactly right. $\beta_2=0.999$ means v_t has a "longer memory" of past gradient magnitudes - it smooths over ~ 1000 steps vs ~ 10 steps for m_t . This gives more stable estimates of gradient variance. ✓

Part 2: The Matrix Structure Problem

Now let's get to the heart of why Muon exists. Remember your answer to Quiz #2: Adam treats all elements independently. Let's see why that's problematic.

2.1: Weight Matrices as Linear Transformations

A weight matrix $\mathbf{W} \in \mathbb{R}^{n \times m}$ doesn't just store $n \times m$ numbers - it represents a **linear map** from $\mathbb{R}^m \rightarrow \mathbb{R}^n$.

When we compute $y = Wx$, we're:

- Taking input vector x (m -dimensional)
- Transforming it to output y (n -dimensional)

The geometry matters! Consider this: which update is "bigger"?

Update A: Add 0.01 to every element of W **Update B:** Replace W with $2W$
(double every element)

Update A changes $n \times m$ elements by 0.01 each. Update B also changes all elements. But Update B **doubles the transformation** - it completely changes how the network behaves! Update A might barely matter.

2.2: Condition Number - Measuring Matrix "Badness"

Let $W = USV^T$ be the **Singular Value Decomposition (SVD)** of W , where:

- U, V are orthogonal matrices (rotations)
- S is diagonal with singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r \geq 0$

[Explanation on how to use condition number to measure matrix condition is very clear and helps me understand the weakness of Adam better!](#)

Condition number: $\kappa(W) = \sigma_1/\sigma_r = (\text{largest singular value})/(\text{smallest singular value})$

Intuitive meaning:

- $\kappa \approx 1$: Matrix is well-conditioned, transforms all directions roughly equally
- $\kappa \gg 1$: Matrix is ill-conditioned, squashes some directions way more than others

Example:

$$W_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \quad \sigma_1=2, \sigma_2=2 \quad \kappa=1 \checkmark \text{ (well-conditioned)}$$

$$W_2 = \begin{bmatrix} 100 & 0 \\ 0 & 1 \end{bmatrix} \quad \sigma_1=100, \sigma_2=1 \quad \kappa=100 \times \text{ (ill-conditioned)}$$

W_2 stretches the x-direction 100× more than the y-direction - it's highly anisotropic.

2.3: Why High Condition Number is Bad for Optimization

When gradient matrices have high condition number, **updates are dominated by just a few directions**.

Empirical observations show that SGD-momentum and Adam updates for 2D parameters in transformers typically have very high condition number - they are almost low-rank matrices, with updates for all neurons dominated by just a few directions.

Visualization:

$$\text{Gradient } G = \begin{bmatrix} 5 & 0.01 \\ 0.01 & 0.01 \end{bmatrix} \quad \sigma_1 \approx 5, \sigma_2 \approx 0.01, \kappa \approx 500$$

Most of the update goes in one direction, ignoring other potentially important directions!

The problem: Adam processes each element independently, so it can't "see" that the gradient is poorly conditioned. It just uses the raw gradient structure.

2.4: What if we could fix the condition number?

Idea: What if, instead of using gradient G directly, we use the "nearest well-conditioned matrix" to G ?

The nearest **orthogonal** matrix ($\kappa=1$, perfectly conditioned) to G is given by its SVD:

If $G = USV^T$, then $\text{Ortho}(G) = UV^T$

This keeps the **direction** (U and V) but sets all singular values to 1!

Quiz #3: Matrix Geometry

Q1: If we have:

$$G = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \quad (\sigma_1=3, \sigma_2=1)$$

What is $\text{Ortho}(G) = UV^T$? (Hint: G is already diagonal, so $U=V=I$)

Q2: For the gradient matrix below, which direction gets most of the update?

$$G = \begin{bmatrix} 100 & 0 \\ 0 & 0.1 \end{bmatrix}$$

Q3: Why might orthogonalizing the gradient (setting all singular values to 1) help with optimization? Think about "rare directions" that are important but have small magnitude in the gradient.

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👤 A1: Ortho(G) is the identity matrix. A2: the first dimension gets most of the update A3: orthogonalizing the gradient let each direction gets similar magnitude of update, so that those important directions with small singular values can still get enough magnitude of update

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Thought: Validated student insights and charted progression toward optimizer concepts.

Excellent! ✓✓✓ All three are spot on!

A1: Correct - $\text{Ortho}([3,0;0,1]) = I$, which has all singular values = 1 ✓

A2: Yes! The first dimension dominates by $1000\times$ ✓

A3: Perfect understanding! You've captured the key insight: orthogonalization effectively increases the scale of "rare directions" which have small magnitude in the update but are nevertheless important for learning. ✓

Part 3: Shampoo - The Precursor to Muon

Now we're ready to understand where Muon comes from! Let's look at the Shampoo optimizer, which pioneered the idea of using matrix structure.

3.1: What is Preconditioning?

In optimization, **preconditioning** means transforming the gradient before using it, to improve the geometry of the optimization landscape.

Basic idea: Instead of $\theta_{t+1} = \theta_t - \eta G$, use $\theta_{t+1} = \theta_t - \eta P G$, where P is a **preconditioner** matrix.

Good preconditioners make the loss landscape more "spherical" (well-conditioned) so optimization is easier.

3.2: The Shampoo Update

Shampoo is defined with preconditioner accumulation, but Bernstein & Newhouse (2024) observed that if preconditioner accumulation is removed, the update becomes:

$$W_{t+1} = W_t - \eta (GG^T)^{-1/4} G (G^T G)^{-1/4}$$

Let's unpack this with SVD. If $G = USV^T$:

$$\begin{aligned} W_{t+1} &= W_t - \eta (US^2U^T)^{-1/4} (USV^T) (VS^2V^T)^{-1/4} \\ &= W_t - \eta (US^{-1/2}U^T) (USV^T) (VS^{-1/2}V^T) \\ &= W_t - \eta U S^{-1/2} S S^{-1/2} V^T \\ &= W_t - \eta UV^T \end{aligned}$$

Amazing insight: The "instantaneous" Shampoo update is exactly the orthogonalized gradient UV^T !

3.3: Why Shampoo is Expensive

To compute $(GG^T)^{-1/4}$ and $(G^T G)^{-1/4}$, Shampoo needs:

1. Matrix multiplications: GG^T and $G^T G$
2. **Inverse fourth roots of matrices** - computationally expensive!

For large matrices (like a 4096×4096 weight in a transformer), this becomes prohibitively slow.

The dream: Get Shampoo's benefits (orthogonalization) without the computational cost.

3.4: Enter Newton-Schulz Iteration

This is where the magic happens! There's a fast algorithm to orthogonalize a matrix without computing SVD or inverse roots.

Newton-Schulz iteration is a sequence of matrix operations that converges to the orthogonal part:

Starting with $\mathbf{X}_0 = \mathbf{G} / \|\mathbf{G}\|$ (normalize), repeatedly apply:

$$\mathbf{X}_{\{k+1\}} = (\mathbf{3I} - \mathbf{X}_k \mathbf{X}_k^T) / 2 \cdot \mathbf{X}_k$$

As $k \rightarrow \infty$, $\mathbf{X}_k \rightarrow \mathbf{UV}^T$ (the orthogonalized matrix)!

Why this is great:

- Only uses matrix multiplications (fast on GPUs!)
 - Can run in bfloat16 (low precision)
 - Converges in ~ 5 iterations for typical neural network gradients
-

Quiz #4: Connecting the Pieces

Q1: Why does Shampoo produce an orthogonalized gradient, even though it's computing complicated inverse fourth roots? (Hint: look at the SVD derivation)

Q2: What's the computational bottleneck of Shampoo that makes it impractical for large models?

Q3: If Newton-Schulz gives us orthogonalization cheaply, why might we want to add momentum *before* orthogonalizing (like Muon does) rather than after?

Think about Q3 carefully - it's subtle but important!

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👤 can you elaborate more on the intuition of how Newton-Schulz iteration works and how can it speed up matrix orthogonalization?

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I think the Newton-Schulz concept is not clear enough so I asked Claude to elaborate more on it

Thought: Deconstructed Newton-Schulz mechanics and orthogonalization acceleration strategies.

Excellent question! Let's dig deep into Newton-Schulz - this is really the key innovation that makes Muon practical.

Deep Dive: Newton-Schulz Iteration

4.1: What Are We Trying to Solve?

Goal: Given matrix G , find the nearest orthogonal matrix O .

For a matrix to be orthogonal: $O^T O = I$ (or $O O^T = I$ depending on shape)

Mathematically: We want $O = \arg \min \|O - G\|$ subject to $O^T O = I$

The solution is $O = UV^T$ where $G = USV^T$ is the SVD.

4.2: Why Not Just Use SVD?

Computing SVD requires:

- **$O(nm^2)$** operations for an $n \times m$ matrix (where $n \geq m$)
- Complex algorithms (QR iterations, divide-and-conquer, etc.)
- **Difficult to parallelize** on GPUs
- **Requires high precision** (usually float32 or float64)

For a 4096×4096 matrix in a transformer, SVD is painfully slow!

4.3: The Newton-Schulz Idea - Fixed Point Iteration

Instead of computing SVD directly, use **iterative refinement**.

Key insight: If X is "close" to orthogonal ($X^T X \approx I$), we can make it "more orthogonal" by applying a correction.

Think of it like gradient descent, but for finding an orthogonal matrix!

4.4: The Classic Newton-Schulz Formula

The basic iteration is:

$$X_{k+1} = X_k (3I - X_k^T X_k) / 2$$

Let's understand why this works:

Intuition via Taylor Expansion

Suppose X_k is close to orthogonal, so $X^T X = I + E$ where E is small.

We want to find a correction that pushes $E \rightarrow 0$.

The matrix inverse square root function: $f(E) = (I + E)^{-1/2}$

For small E , Taylor expansion gives: $(I + E)^{-1/2} \approx I - E/2 + (3E^2)/8 + \dots$

The Newton-Schulz iteration implements a polynomial approximation to this!

Why $(3I - X^T X)/2$?

Let's check: if $X^T X = I + E$ (nearly orthogonal):

$$\begin{aligned} X_{k+1} &= X_k (3I - (I + E))/2 \\ &= X_k (2I - E)/2 \\ &= X_k (I - E/2) \end{aligned}$$

This applies a correction that reduces the error E !

4.5: Muon's Enhanced Version - Using Higher Order Polynomials

Muon uses a **quintic (degree 5) polynomial** for faster convergence:

$$X_{k+1} = (aI + b(X_k X_k^T) + c(X_k X_k^T)^2) X_k$$

where $a=3.4445$, $b=-4.7750$, $c=2.0315$ (tuned coefficients)

Let's see why this is better:

From Iteration to Polynomial

Remember $G = USV^T$. After normalization: $G/\|G\|$ has singular values in $[0,1]$.

Each Newton-Schulz step applies a polynomial ϕ to the singular values:

$$X_k = U \phi^k(S) V^T$$

where ϕ^k means "apply ϕ k times"

For Muon: $\varphi(\mathbf{x}) = \mathbf{ax} + \mathbf{bx}^3 + \mathbf{cx}^5$

The goal: make $\varphi^5(x) \approx 1$ for all $x \in [0,1]$, so after 5 iterations, all singular values $\rightarrow 1$!

4.6: Visual Intuition

Let me show you how the polynomial works:

Initial singular values: $[0.1, 0.3, 0.5, 0.8, 1.0]$

After 1 iteration:

$\varphi([0.1, 0.3, 0.5, 0.8, 1.0]) \rightarrow [0.35, 0.67, 0.86, 1.02, 1.03]$

After 2 iterations:

$\varphi^2([0.1, 0.3, 0.5, 0.8, 1.0]) \rightarrow [0.70, 0.95, 1.01, 1.02, 1.01]$

After 3 iterations:

$\varphi^3([0.1, 0.3, 0.5, 0.8, 1.0]) \rightarrow [0.94, 1.00, 1.00, 1.00, 1.00]$

All singular values $\rightarrow 1$! Matrix is now orthogonal!

The magic: Small singular values grow faster than large ones (note $\varphi'(0) = a = 3.4445$ is large), so everything converges to 1 quickly.

4.7: Why Newton-Schulz is Fast

Comparison table:

Method	Operations per step	GPU-friendly?	Precision needed	Convergence
SVD	$O(nm^2)$ complex	❌ No	High (fp32/fp64)	1 step
Coupled Newton	$O(m^3)$ per iter	❌ No	High (fp32)	~10 iters
Newton-Schulz	$O(nm^2)$ simple	✅ Yes	Low (bf16)	~5 iters

Key advantages:

1. **Matrix multiplications only:** Modern GPUs are optimized for this (NVIDIA's tensor cores!)
2. **Low precision:** Works in bfloat16, which is 2x faster than float32
3. **Highly parallelizable:** Each element computed independently
4. **Numerically stable:** The polynomial keeps values bounded

4.8: Computational Cost Analysis

For an $n \times m$ matrix ($n \geq m$), each Newton-Schulz step requires:

- **$2nm^2$ FLOPs** for $X @ X^T$
- **$2nm^2$ FLOPs** for (stuff) @ X
- **$2m^3$ FLOPs** for matrix-matrix products involving $m \times m$ matrices

Total: $4nm^2 + 2m^3$ FLOPs per iteration

With 5 iterations: **$20nm^2 + 10m^3$ FLOPs**

But: In a training step, we do forward + backward pass = $6nmB$ FLOPs (B = batch size)

Overhead ratio: $(20nm^2 + 10m^3)/(6nmB) = \sim(3.3m/B + 1.7m^3/nmB)$



For transformers with large $B \gg m$, **this is typically $< 1\%$** !

Quiz #5: Newton-Schulz Understanding

Q1: Why can Newton-Schulz run in bfloat16 while SVD needs float32? (Hint: think about what happens if there's small numerical error at each step)

Q2: Look at this convergence:

Singular values: $[0.01, 0.5, 1.0]$
After φ : $[0.10, 0.86, 1.03]$

Why does $0.01 \rightarrow 0.10$ ($10\times$ increase) while $1.0 \rightarrow 1.03$ (only 3% increase)?

Q3: For a 2048×2048 weight matrix with batch size $B=524288$ tokens, what's the approximate FLOP overhead of 5 Newton-Schulz iterations? (Use the formula above)

Take your time - Q3 involves some calculation!

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👤 I still don't get a good intuition about how Newton-Schulz work to do the diagonalization, maybe can you walk me through an example?

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Thought: Devised numerical walkthrough demonstrating Newton-Schulz orthogonalization mechanics.

Excellent question! Abstract explanations are hard - let's work through a concrete numerical example step by step.

Worked Example: Newton-Schulz in Action

Step 0: Starting Matrix

Let's take a simple 2×2 matrix:

$$G = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

SVD of G:

- $U = I$ (identity, no rotation)
- $S = [3, 1]$ (singular values)
- $V = I$ (identity, no rotation)

So $G = USV^T = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$

Our goal: Transform $G \rightarrow UV^T = [1, 0; 0, 1] = I$ (all singular values become 1)

Step 1: Normalize

First, normalize G by its Frobenius norm:

$$\|G\|_F = \sqrt{3^2 + 0^2 + 0^2 + 1^2} = \sqrt{10} \approx 3.162$$

$$X_0 = G/\|G\|_F = [3/3.162, 0; 0, 1/3.162] \approx [0.949, 0; 0, 0.316]$$

Now X_0 has singular values: $S_0 = [0.949, 0.316]$ (scaled version)

Check orthogonality:

$$X_0^T X_0 = \begin{bmatrix} 0.949^2 & 0 \\ 0 & 0.316^2 \end{bmatrix} = \begin{bmatrix} 0.901 & 0 \\ 0 & 0.100 \end{bmatrix}$$

Not orthogonal! Diagonal should be $[1, 1]$.

Step 2: First Newton-Schulz Iteration

We'll use the simple formula first: $X_{k+1} = X_k(3I - X_k^T X_k)/2$

Compute $X_0^T X_0$:

$$X_0^T X_0 = \begin{bmatrix} 0.901 & 0 \\ 0 & 0.100 \end{bmatrix}$$

Compute $3I - X_0^T X_0$:

$$3\mathbf{I} - \mathbf{X}_0^T \mathbf{X}_0 = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} 0.901 & 0 \\ 0 & 0.100 \end{bmatrix} = \begin{bmatrix} 2.099 & 0 \\ 0 & 2.900 \end{bmatrix}$$

Compute $(3\mathbf{I} - \mathbf{X}_0^T \mathbf{X}_0)/2$:

$$= \begin{bmatrix} 1.050 & 0 \\ 0 & 1.450 \end{bmatrix}$$

Compute $\mathbf{X}_1 = \mathbf{X}_0 \cdot (3\mathbf{I} - \mathbf{X}_0^T \mathbf{X}_0)/2$:

$$\begin{aligned} \mathbf{X}_1 &= \begin{bmatrix} 0.949 & 0 \\ 0 & 0.316 \end{bmatrix} \cdot \begin{bmatrix} 1.050 & 0 \\ 0 & 1.450 \end{bmatrix} \\ &= \begin{bmatrix} 0.949 \times 1.050 & 0 \\ 0 & 0.316 \times 1.450 \end{bmatrix} \\ &= \begin{bmatrix} 0.996 & 0 \\ 0 & 0.458 \end{bmatrix} \end{aligned}$$

New singular values: $\mathbf{S}_1 \approx [0.996, 0.458]$

Notice:

- $0.949 \rightarrow 0.996$ (got closer to 1!)
- $0.316 \rightarrow 0.458$ (moved toward 1!)

Step 3: Second Iteration

Compute $\mathbf{X}_1^T \mathbf{X}_1$:



$$X_1^T X_1 = \begin{bmatrix} 0.992 & 0 \\ 0 & 0.210 \end{bmatrix}$$

Compute $3I - X_1^T X_1$:

$$= \begin{bmatrix} 2.008 & 0 \\ 0 & 2.790 \end{bmatrix}$$

Compute $X_2 = X_1 \cdot (3I - X_1^T X_1)/2$:

$$\begin{aligned} X_2 &= \begin{bmatrix} 0.996 & 0 \\ 0 & 0.458 \end{bmatrix} \cdot \begin{bmatrix} 1.004 & 0 \\ 0 & 1.395 \end{bmatrix} \\ &= \begin{bmatrix} 1.000 & 0 \\ 0 & 0.639 \end{bmatrix} \end{aligned}$$

New singular values: $S_2 \approx [1.000, 0.639]$

Great progress! First singular value is already at 1.000!

Step 4: Third Iteration

Compute $X_2^T X_2$:

$$X_2^T X_2 = \begin{bmatrix} 1.000 & 0 \\ 0 & 0.408 \end{bmatrix}$$

Compute X_3 :

$$X_3 = \begin{bmatrix} 1.000 & 0 \\ 0 & 0.639 \end{bmatrix} \cdot \begin{bmatrix} 1.000 & 0 \\ 0 & 1.296 \end{bmatrix}$$

$$= \begin{bmatrix} 1.000 & 0 \\ 0 & 0.828 \end{bmatrix}$$

Singular values: $S_3 \approx [1.000, 0.828]$

Step 5: Fourth Iteration

Continuing the pattern:

$$X_4 \approx [1.000, 0; 0, 0.938]$$

Singular values: $S_4 \approx [1.000, 0.938]$

Step 6: Fifth Iteration

$$X_5 \approx [1.000, 0; 0, 0.984]$$

Singular values: $S_5 \approx [1.000, 0.984]$

Almost there!

Convergence Summary

Watch how the singular values evolve:

Iteration	σ_1 (large)	σ_2 (small)
Initial	0.949	0.316
After 1	0.996	0.458
After 2	1.000	0.639
After 3	1.000	0.828
After 4	1.000	0.938
After 5	1.000	0.984
Target	1.000	1.000

Key Insights from This Example

1. The "Correction Factor" Pattern

At each step, the correction $(3I - X^T X)/2$ does this:

- For σ close to 1: correction ≈ 1 (small change)
- For σ far from 1: correction > 1 (large change)

Example from iteration 1:

- $\sigma_1 = 0.949$ (close to 1) $\rightarrow$ correction = 1.050 $\rightarrow$ new $\sigma = 0.996$
- $\sigma_2 = 0.316$ (far from 1) $\rightarrow$ correction = 1.450 $\rightarrow$ new $\sigma = 0.458$

The smaller value gets a **bigger boost**!

2. Why Small Singular Values Grow Faster

Look at the correction factor formula:

$$(3I - \sigma^2 I)/2 = (3 - \sigma^2)/2$$

If $\sigma = 0.1$: correction = $(3 - 0.01)/2 = 1.495$ (50% boost!)
If $\sigma = 0.9$: correction = $(3 - 0.81)/2 = 1.095$ (10% boost)
If $\sigma = 1.0$: correction = $(3 - 1.00)/2 = 1.000$ (no change)

This is **self-correcting**: values far from 1 get pushed harder toward 1!

3. Geometric Interpretation

Think of singular values as "stretching factors" in different directions:

Original: Stretches x-axis 3×, y-axis 1× (unbalanced!)
After NS: Stretches x-axis 1×, y-axis 1× (balanced!)

Newton-Schulz "redistributes" the stretching evenly across all directions.

With Muon's Quintic Polynomial

Muon uses $\varphi(\mathbf{x}) = \mathbf{ax} + \mathbf{bx}^3 + \mathbf{cx}^5$ with optimized coefficients.

Let's see the difference on our small singular value $\sigma = 0.316$:

Classic NS: $0.316 \rightarrow 0.458 \rightarrow 0.639 \rightarrow 0.828 \rightarrow 0.938$ (5 steps)

Muon's φ (approximate): $0.316 \rightarrow 0.65 \rightarrow 0.92 \rightarrow 0.99 \rightarrow 1.00$ (converges faster!)

The quintic polynomial is tuned to push small values toward 1 more aggressively (recall $a=3.4445$ is large).

Quiz #6: Understanding the Convergence

Q1: In our example, why did $\sigma_1=0.949$ reach 1.000 by iteration 2, while $\sigma_2=0.316$ needed 5+ iterations?

Q2: What would happen if we started with a singular value $\sigma=1.5$ (greater than 1)? Will Newton-Schulz bring it back down to 1? (Hint: compute the correction factor)

Q3: Why do we normalize $X_0 = G/\|G\|$ at the start? What would go wrong if we didn't?

Think about these, especially Q2 and Q3!

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⊙ A1: because smaller values would get push to 1 more aggresively A2: No, it will not. A3: To avoid getting too large singular value, which wil eventually explode instead of converge to 1 after applying the Newton-Schulz iteration.

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Thought: Verified student answers, identified computational error, and clarified convergence requirements.

Good thinking! Let me check your answers:

A1: ✓ Correct! You've got the key insight - values further from 1 get pushed more aggressively. Perfect!

A2: ✗ Not quite! Let's calculate:

If $\sigma = 1.5$:

Correction = $(3 - \sigma^2)/2 = (3 - 2.25)/2 = 0.375$
New $\sigma = 1.5 \times 0.375 = 0.5625$

So it **does** bring it down! But notice: $1.5 \rightarrow 0.5625$ overshoots below 1, then would bounce back up. The iteration might oscillate rather than converge smoothly.

This is exactly why normalization matters!

A3: ✓ Mostly correct! You have the right intuition. More precisely:

The polynomial $\phi(x)$ is designed to converge to 1 for $x \in [0, 1]$. By normalizing $G \rightarrow G/\|G\|$, we guarantee all singular values are in $[0, 1]$, which is the **convergence domain** of the polynomial.

Without normalization:

- If $\sigma > 1$: might oscillate or behave unpredictably

- If $\sigma \gg 1$: polynomial not designed for this regime

The normalization is mathematically "free" because $\text{Ortho}(cG) = \text{Ortho}(G)$ for any scalar c .

Part 5: Putting It All Together - MUON

Now you understand all the pieces! Let's see how they combine into Muon.

5.1: The Muon Algorithm

Muon = Momentum Orthogonalized by Newton-schulz

Here's the complete algorithm:

```
1. Compute gradient:  $G_t = \nabla L(W_t)$ 

2. Apply momentum:
    $V_t = \beta V_{t-1} + G_t$  ( $\beta = 0.95$ )

3. Orthogonalize with Newton-Schulz:
    $V_{t\_ortho} = \text{NewtonSchulz5}(V_t)$ 

4. Update weights:
    $W_{t+1} = W_t - \eta V_{t\_ortho}$ 
```

Key design choices:

- Momentum **before** orthogonalization (not after!)
- Use Nesterov momentum (slightly better empirically)

- Apply only to 2D parameters (weight matrices)
- Use AdamW for 1D parameters, embeddings, and output layers

5.2: Why Momentum Before Orthogonalization?

This is subtle but important! Two options:

Option A (Orthogonal-SGDM): Orthogonalize → then momentum

```
G_ortho = Ortho(G_t)
V_t =  $\beta$  V_{t-1} + G_ortho
Update:  $W \leftarrow W + \eta V_t$ 
```

It's able to point out the subtle design of Muon to perform momentum before orthogonalization (and gives eligible reasons to justify this design)

Option B (Muon): Momentum → then orthogonalize

```
V_t =  $\beta$  V_{t-1} + G_t
V_ortho = Ortho(V_t)
Update:  $W \leftarrow W + \eta V_ortho$ 
```

Why is Option B (Muon) better?

Think about what momentum does: it accumulates gradients over time, building up signal in consistent directions.

- **Option A:** We throw away magnitude info at each step (via Ortho), then accumulate. We're accumulating **directions only**.
- **Option B:** We accumulate gradients with their magnitudes, building up strong signal. **Then** we orthogonalize the accumulated update. The orthogonalization acts on a richer signal that incorporates history!

Empirical result: Muon performs better than Orthogonal-SGDM, which was outperformed by standard SGD-Momentum in experiments.



5.3: Muon's Architecture-Specific Behavior

Not all parameters get the Muon treatment:

Parameter Type	Optimizer	Why?
Hidden layer weights (2D)	Muon	Matrix structure matters!
Embeddings (2D)	AdamW	Input layer needs different dynamics
Output/classifier (2D)	AdamW	Empirically works better
Layer norms (1D)	AdamW	Not matrices
Biases (1D)	AdamW	Not matrices

For transformers: Apply Muon to Q, K, V parameters separately rather than as one combined QKV matrix.

5.4: Muon vs AdamW - Key Differences

Aspect	AdamW	Muon
View of parameters	Flat vector of scalars	Structured matrices
Adaptivity	Per-element via <code>v_t</code>	Per-matrix via orthogonalization
Memory	2× params (<code>m_t</code> , <code>v_t</code>)	1× params (<code>v_t</code> only)
Computation	Cheap elementwise ops	Newton-Schulz iterations (~1% overhead)
Learning rate transfer	Must retune for different widths	Automatic across widths!

5.5: The Learning Rate Transfer Magic

This is one of Muon's most remarkable properties!

Problem with AdamW: When you change model width (e.g., 512 → 1024 hidden dim), you need to retune the learning rate.

Muon's solution: By exploiting the geometry of matrices as linear operators between RMS-normed vector spaces, Muon achieves automatic learning rate transfer across different model widths.

Why? The orthogonalization normalizes updates by their matrix structure, making the effective step size invariant to width changes!

Practical impact: Train a small model, scale up, and the same hyperparameters work! Huge time/money saver.

5.6: Empirical Results

Muon has achieved impressive results:

- Improved CIFAR-10 training record from 3.3 to 2.6 A100-seconds (94% accuracy)
- 1.35× speedup on NanoGPT speedrunning task
- Trained 1.5B parameter transformer to GPT-2 XL performance in 10 H100-hours vs 13.3 hours with AdamW

Recent work shows Muon achieves ~2× computational efficiency compared to AdamW in compute-optimal training, and has been used to train models up to 1 trillion parameters.

Final Conceptual Quiz #7:

Q1: Why does Muon use AdamW for embeddings and output layers instead of applying orthogonalization? What's fundamentally different about these layers?

Q2: Suppose you're training two models:

- Model A: 768-dimensional hidden layers
- Model B: 3072-dimensional hidden layers (4× wider)

With AdamW, you'd likely need to adjust learning rates. Why would Muon work with the same LR for both?

Q3: A researcher proposes "MuonPlus": apply orthogonalization to ALL parameters including 1D. Why might this be a bad idea?

These questions test your deep understanding of Muon's design philosophy!

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👤 A1: cause embedding layers would have sparse gradients A2: cause Muon updates could preserve the RMS-to-RMS norm to the order of 1, so that it's invariant to the width of the weight matrix Q3: Newton-schulz is a matrix based algorithm which is not defined for 1D vector

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Thought: Evaluated three student answers, validating insights while identifying incomplete reasoning.

Excellent answers! Let me give you detailed feedback:

A1: ✓ Partially correct! Sparse gradients are indeed relevant for embeddings! But there's more to the story:

From the research: The optimization dynamics of embedding layers being different follows from modular norm theory, while output layers needing different treatment is driven by empirics.

So it's a combination of:

- **Theoretical:** Embeddings have different geometric properties (input layer in a "modular" architecture)
- **Practical:** Sparse gradients (only tokens in batch get updates)
- **Empirical:** Output layers just work better with AdamW (discovered through experiments)

Your sparse gradient intuition is spot on though! ✓

A2: ✓✓ Excellent! This is exactly right! You've grasped the deep theoretical reason. Muon exploits the geometry of matrices as linear operators between RMS-normed vector spaces, which provides learning rate transfer. The orthogonalization makes the update's operator norm invariant to width scaling. This is sophisticated understanding! ✓✓

A3: ✓ Perfect! Yes! Newton-Schulz requires computing $X^T X$ or XX^T to iteratively reach orthogonality. For a 1D vector v , $v^T v$ is just a scalar $\|v\|^2$, and "orthogonalizing" a scalar makes no sense. The whole concept of condition number, singular values, etc. requires at least 2D structure. ✓

Part 6: The Complete Picture

You've got the fundamentals down! Let me give you the full Muon story:

6.1: Muon's Position in Optimization History

SGD (1950s)

↓

SGD + Momentum (1960s)

↓

Adaptive methods: AdaGrad, RMSprop (2010s)

↓

Adam/AdamW (2014-2017) ← Current standard

↓

Second-order methods: Shampoo (2018)

↓ (too expensive!)

↓

Newton-Schulz speedup (Bernstein & Newhouse, 2024)

↓

Muon (Jordan et al., 2024) ← We are here!

It fails to recover muP at all

6.2: Why Muon Matters - The Three Pillars

Pillar 1: Efficiency

- **Sample efficiency:** Reaches target loss in fewer gradient steps
- **Compute efficiency:** <1% overhead compared to SGD
- **Wall-clock efficiency:** Faster training time than AdamW

Pillar 2: Scalability

- Works from 124M to 1.5B+ parameters
- Scaled to 1 trillion parameter models (KIMI Moonshot)
- Learning rate transfers across model sizes

Pillar 3: Simplicity

- Only 3 hyperparameters: learning rate, momentum (0.95), weight decay
- No need for complex tuning
- Default settings work well

6.3: The Core Innovation Summary

Traditional optimizers (Adam):

- Element-wise thinking
- Maintain per-parameter statistics
- High memory cost (2× parameters)

Muon:

- Matrix-aware thinking
- Exploit geometric structure via orthogonalization
- Lower memory cost (1× parameters)
- Respects the "rare directions" in gradient space



Key equation:

```
Traditional:  $W \leftarrow \eta * (\text{momentum of gradients})$   
Muon:       $W \leftarrow \eta * \text{Ortho}(\text{momentum of gradients})$ 
```

That one extra step (orthogonalization) changes everything!

6.4: When to Use Muon

✅ Good for:

- Training transformers from scratch (pretraining)
- Models with large 2D weight matrices
- When you need to scale across model sizes
- Budget-constrained training

? Unknown/Under research:

- Finetuning (may need AdamW)
 - Reinforcement learning
 - Very small models (<10M parameters)
 - Extreme sparsity scenarios
-

6.5: Practical Implementation Tips

From the actual code:

```
# Muon for 2D parameters (hidden layers)  
muon = Muon(
```

```

hidden_params,
lr=0.01,          # Learning rate
momentum=0.95,    # Momentum coefficient
nesterov=True,    # Use Nesterov momentum
ns_steps=5        # Newton-Schulz iterations
)

# AdamW for everything else
adamw = AdamW(
    [embeddings, output_layer, layer_norms, biases],
    lr=0.001,
    weight_decay=0.01
)

# Training loop
for batch in dataloader:
    loss = forward(batch)
    loss.backward()

    muon.step()    # Updates hidden layers
    adamw.step()   # Updates everything else

    muon.zero_grad()
    adamw.zero_grad()

```

6.6: The Evidence Standard

Muon's success in competitive tasks (NanoGPT speedrunning, CIFAR-10) provides strong evidence because: the baseline is already well-tuned, and if Muon weren't better, someone would have removed it and set a new record with standard AdamW (which has less overhead).

This is a key methodological insight: competitive benchmarks solve the "undertuned baseline" problem that plagues optimization research!

Final Practical Quiz #8:

You're now ready for a real-world scenario!

Scenario: You're training a 774M parameter GPT-style model on a custom dataset. Currently using AdamW with:

- LR = $3e-4$
- Batch size = 256
- Model dim = 2048
- Training taking 48 hours on 8xA100

Questions:

Q1: Which parameters in your model should use Muon vs AdamW? List specific layer types.

Q2: Estimate the FLOP overhead of adding Muon. Use the formula: overhead $\approx 5m/B$ where $m=2048$ and $B=\text{batch_size_in_tokens}$. If sequence length is 2048, what's B in tokens?

Q3: If Muon gives $1.3\times$ sample efficiency speedup (like in NanoGPT), but adds 0.5% wall-clock overhead, what's your expected total training time? Will it be faster?

Q4: You want to scale to a 3B parameter model ($m=4096$). With AdamW, you'd normally retune the LR. Why might Muon let you keep the same LR?

Take your time with these - they're testing your ability to apply Muon knowledge!

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